

Dr. Daegeun Jo

Postdoctoral Researcher

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EDUCATION

Ph.D. in Physics

Department of Physics, Pohang University of Science and Technology

📅 Aug. 2023

📍 Pohang, Republic of Korea

- Supervisor: Prof. Hyun-Woo Lee
- Thesis: Generation, Detection, and Application of Orbital Dynamics

B.S. in Physics

Department of Physics, Hanyang University

📅 Feb. 2017

📍 Seoul, Republic of Korea

RESEARCH EXPERIENCE

Postdoctoral Researcher

Department of Physics and Astronomy, Uppsala University

📅 Sep. 1. 2023 — Present

📍 Uppsala, Sweden

- Supervisor: Prof. Peter M. Oppeneer
- Includes parental leave (May-Jul. 2025)

Guest Scientist

Peter Grünberg Institut and Institute for Advanced Simulation, Forschungszentrum Jülich

📅 Apr. 18. 2022 — Jul. 16. 2022

📍 Jülich, Germany

- Group of Prof. Yuriy Mokrousov

SKILLS

Theoretical calculation for orbitronics and spintronics

- Electronic properties of materials
- Current-induced orbital and spin responses (orbital/spin Hall effect, orbital/spin Edelstein effect)
- Magneto-optic effects
- Orbital torque and spin-orbit torque in magnetic heterostructures

Tools

- Tight-binding model (empirical or semi-realistic)
- First-principles calculation (Quantum ESPRESSO, FLEUR) with Wannier90 code
- Programming languages: Python and Fortran
- High-performance computing with MPI

Languages

- Korean, English

OTHER EXPERIENCE

Teaching Assistant

Pohang University of Science and Technology

📅 2017 – 2018

📍 Pohang, Republic of Korea

- 2018. Analytical Mechanics for graduate students
- 2017. General Physics II for undergraduate students
- 2017. General Physics Lab. I for undergraduate students

Compulsory Military Service

Weather Group of Republic of Korea Air Force

📅 2012 – 2014

📍 Republic of Korea

- Discharged at the rank of sergeant

AWARDS & HONORS

- 2024. S-Oil Dissertation of the Year Award (S-OIL Science and Culture Foundation)
- 2023. Best Paper Award for Graduate Students in the Department of Physics (POSTECH)
- 2023. B. I. Min Best Condensed Matter Theory Paper Award (POSTECH)
- 2021. Silver Prize at the 27th Samsung Humantech Paper Award (Samsung Electronics Co., Ltd.)
- 2018. Best presentation award (poster) at 2018 Korean Physical Society Fall Meeting
- 2018–2023. Global Ph.D. Fellowship (funded by National Research Foundation)
- 2011. National Science & Technology Scholarship (funded by Korea Student Aid Foundation)

SELECTED PUBLICATIONS

†: These authors contributed equally.

- S. Han[†], **D. Jo**[†], I. Baek[†], S. Cheon, P. M. Oppeneer, and H.-W. Lee, “Harnessing Magnetic Octupole Hall Effect to Induce Torque in Altermagnets”, [Phys. Rev. Lett. **135**, 076705 \(2025\)](#).
- **D. Jo**, D. Go, Y. Mokrousov, P. M. Oppeneer, S.-W. Cheong, and H.-W. Lee, “Weak Ferromagnetism in Altermagnets from Alternating g -Tensor Anisotropy”, [Phys. Rev. Lett. **134**, 196703 \(2025\)](#).
- Y.-G. Choi[†], **D. Jo**[†], K.-H. Ko[†], D. Go, K.-H. Kim, H. G. Park, C. Kim, B.-C. Min, G.-M. Choi, and H.-W. Lee, “Observation of the orbital Hall effect in a light metal Ti”, [Nature **619**, 52–56 \(2023\)](#).
- **D. Jo**, D. Go, and H.-W. Lee, “Gigantic intrinsic orbital Hall effects in weakly spin-orbit coupled metals”, [Phys. Rev. B **98**, 214405 \(2018\)](#).

FULL LIST OF PUBLICATIONS

†: These authors contributed equally.

Peer-reviewed journals

- K.-H. Ko[†], **D. Jo**[†], S. Kim, P. Oppeneer, H.-W. Lee, and G.-M. Choi, “Electron scattering effect on orbital transport in highly disordered nanocrystalline Mn”, [Nano Letters, accepted for publication](#).
- **D. Jo** and P. M. Oppeneer, “Unconventional orbital currents and torques due to ferro-rotational orbital textures”, [npj Spintronics **4**, 22 \(2026\)](#).
- K.-H. Ko[†], **D. Jo**[†], P. M. Oppeneer, H.-W. Lee, and G.-M. Choi, “Magneto-optical observation of electrically generated orbital polarization in pristine Cu and oxidized Cu”, [Communications Physics **9**, 174 \(2026\)](#).
- S. Alikhah, **D. Jo**, M. Berritta, and P. M. Oppeneer, “Theory for magneto-optical detection of the interfacial orbital Rashba-Edelstein effect”, [Communications Physics **9**, 131 \(2026\)](#).

- **D. Jo** and P. M. Oppeneer, “Theoretical study of orbital torque: Dependence on ferromagnet species and nonmagnetic layer thickness”, *Journal of Applied Physics* **139**, 103907 (2026).
- K. Du, **D. Jo**, X. Xu, F.-T. Huang, M.-H. Lee, M.-W. Chu, K. Wang, X. Guo, L. Zhao, D. Vanderbilt, H.-W. Lee, and S.-W. Cheong, “Electric toroidal invariance generates distinct transverse electromagnetic responses”, *Nature Physics* **22**, 61–67 (2026).
- H. Hayashi, J. Chen, **D. Jo**, S. Sakamoto, T. Gao, D. Go, Y. Mokrousov, H.-W. Lee, S. Miwa, and K. Ando, “Crystallographic Engineering for Enhanced Orbital Torque”, *Nano Letters* **25**, 15124–15129 (2025).
- S. Han[†], **D. Jo**[†], I. Baek[†], S. Cheon, P. M. Oppeneer, and H.-W. Lee, “Harnessing Magnetic Octupole Hall Effect to Induce Torque in Altermagnets”, *Phys. Rev. Lett.* **135**, 076705 (2025).
- **D. Jo**, D. Go, Y. Mokrousov, P. M. Oppeneer, S.-W. Cheong, and H.-W. Lee, “Weak Ferromagnetism in Altermagnets from Alternating g -Tensor Anisotropy”, *Phys. Rev. Lett.* **134**, 196703 (2025).
- **D. Jo**, D. Go, G.-M. Choi, and H.-W. Lee, “Spintronics meets orbitronics: Emergence of orbital angular momentum in solids”, *npj Spintronics* **2**, 19 (2024).
- H. Moriya[†], M. Taniguchi[†], **D. Jo**[†], D. Go, N. Soya, H. Hayashi, Y. Mokrousov, H.-W. Lee, and K. Ando, “Observation of Long-Range Current-Induced Torque in Ni/Pt Bilayers”, *Nano Letters* **24**, 6459–6464 (2024).
- J. Kim, J. Uzuhashi, M. Horio, T. Senoo, D. Go, **D. Jo**, T. Sumi, T. Wada, I. Matsuda, T. Ohkubo, S. Mitani, H.-W. Lee, and Y. Otani, “Oxide layer dependent orbital torque efficiency in ferromagnet/Cu/oxide heterostructures”, *Phys. Rev. Mater.* **7**, L111401 (2023).
- Y.-G. Choi[†], **D. Jo**[†], K.-H. Ko[†], D. Go, K.-H. Kim, H. G. Park, C. Kim, B.-C. Min, G.-M. Choi, and H.-W. Lee, “Observation of the orbital Hall effect in a light metal Ti”, *Nature* **619**, 52–56 (2023).
- D. Go, **D. Jo**, K.-W. Kim, S. Lee, M.-G. Kang, B.-G. Park, S. Blügel, H.-W. Lee, and Y. Mokrousov, “Long-Range Orbital Torque by Momentum-Space Hotspots”, *Phys. Rev. Lett.* **130**, 246701 (2023).
- H. Hayashi, **D. Jo**, D. Go, Y. Mokrousov, H.-W. Lee, and K. Ando, “Observation of long-range orbital transport and giant orbital torque”, *Communications Physics* **6**, 32 (2023).
- D. Lee, D. Go, H.-J. Park, W. Jeong, H.-W. Ko, D. Yun, **D. Jo**, S. Lee, G. Go, J. H. Oh, K.-J. Kim, B.-G. Park, B.-C. Min, H. C. Koo, H.-W. Lee, O. Lee, and K.-J. Lee, “Orbital torque in magnetic bilayers”, *Nature Communications* **12**, 6710 (2021).
- D. Go, **D. Jo**, H.-W. Lee, M. Kläui, and Y. Mokrousov, “Orbitronics: Orbital currents in solids”, *EPL (Europhysics Letters)* **135**, 37001 (2021).
- D. Go, **D. Jo**, T. Gao, K. Ando, S. Blügel, H.-W. Lee, and Y. Mokrousov, “Orbital Rashba effect in a surface-oxidized Cu film”, *Phys. Rev. B* **103**, L121113 (2021).
- K. Zhang, S. Han, Y. Lee, M. J. Coak, J. Kim, I. Hwang, S. Son, J. Shin, M. Lim, **D. Jo**, K. Kim, D. Kim, H.-W. Lee, and J.-G. Park, “Gigantic Current Control of Coercive Field and Magnetic Memory Based on Nanometer-Thin Ferromagnetic van der Waals Fe₃GeTe₂”, *Advanced Materials* **33**, 2004110 (2021).
- J. Kim, D. Go, H. Tsai, **D. Jo**, K. Kondou, H.-W. Lee, and Y. Otani, “Nontrivial torque generation by orbital angular momentum injection in ferromagnetic-metal/Cu/Al₂O₃ trilayers”, *Phys. Rev. B* **103**, L020407 (2021).
- Z. C. Zheng, Q. X. Guo, **D. Jo**, D. Go, L. H. Wang, H. C. Chen, W. Yin, X. M. Wang, G. H. Yu, W. He, H.-W. Lee, J. Teng, and T. Zhu, “Magnetization switching driven by current-induced torque from weakly spin-orbit coupled Zr”, *Phys. Rev. Research* **2**, 013127 (2020).
- **D. Jo**, D. Go, and H.-W. Lee, “Gigantic intrinsic orbital Hall effects in weakly spin-orbit coupled metals”, *Phys. Rev. B* **98**, 214405 (2018).
- D. Go, **D. Jo**, C. Kim, and H.-W. Lee, “Intrinsic Spin and Orbital Hall Effects from Orbital Texture”, *Phys. Rev. Lett.* **121**, 086602 (2018).

Preprints

- Y. Masuda, T. Hirai, **D. Jo**, N. Yano, P. M. Oppeneer, H. Sepehri-Amin, K.-i. Uchida, and K. Ando, *Observation of the orbital Nernst effect*, 2026, [arXiv:2609.27604 \[cond-mat.mes-hall\]](#).
- **D. Jo**, P. M. Oppeneer, and M. Yarmohammadi, *Interplay of nonrelativistic and relativistic spin splittings in altermagnets*, 2026, [arXiv:2608.20528 \[cond-mat.mtrl-sci\]](#).
- U. Shashank, A. Kumar, **D. Jo**, T. N. A. Nguyen, J.-G. Choi, S. Ghosh, M. Strach, L. Zeng, A. B. Yankovich, R. Khymyn, A. A. Awad, E. Olsson, P. M. Oppeneer, and J. Åkerman, *Spin-Orbital Hall Nano-Oscillators using PtCr/NiFe*, 2026, [arXiv:2608.08504 \[cond-mat.mes-hall\]](#).
- M. Yarmohammadi, **D. Jo**, M. Berritta, L. Šmejkal, J. K. Freericks, and P. M. Oppeneer, *Giant perpendicular Edelstein polarization in 2D compensated magnets via bichromatic Floquet driving*, 2026, [arXiv:2606.31867 \[cond-mat.mtrl-sci\]](#).

- N. Vijayan, D. Kumar, A. Du, M. Sastges, L. Gao, Z. Xiao, D. Go, J. O. Ledesma-Martin, H. I. Wang, **D. Jo**, P. M. Oppeneer, R. Gupta, G. Jakob, S. Krishnia, Y. Mokrousov, and M. Kläui, *Itinerant Orbital Hall Effect Mechanism Leading to Large Negative Orbital Torques from Light Metal Vanadium*, 2025, [arXiv:2508.16339](https://arxiv.org/abs/2508.16339) [[cond-mat.mes-hall](https://arxiv.org/archive/cond-mat)].
- J. C. Idrobo, J. Rusz, G. Datt, **D. Jo**, S. Alikhah, D. Muradas, U. Nounbe, M. V. Kamalakar, and P. M. Oppeneer, *Direct observation of nanometer-scale orbital angular momentum accumulation*, 2024, [arXiv:2403.09269v1](https://arxiv.org/abs/2403.09269v1) [[cond-mat.mes-hall](https://arxiv.org/archive/cond-mat)].
- J. Kim, J. Uzuhashi, D. Go, **D. Jo**, T. Ohkubo, S. Mitani, H.-W. Lee, and Y. Otani, *Emerging effects of oxygen accumulation on orbital torque*, preprint available at Research Square, 2023.

PRESENTATIONS

Oral

- "Unconventional orbital currents and torques due to ferro-rotational orbital textures", Orbitronics Workshop 2026, Carry-le-Rouet, France, May 5–7, 2026 (Contributed talk)
- "Weak orbital ferromagnetism in altermagnets from g -tensor anisotropy", Sweden Magnetism Meet 2026 (SMM-2026), Gothenburg, Sweden, Jun. 1–5, 2026 (Contributed talk)
- "Unconventional orbital currents in ferro-rotational systems", Swedish-Korean Orbitronics Workshop 2025, Gothenburg, Sweden, Jul. 17–18, 2025 (Contributed talk)
- "Unconventional orbital currents in ferro-rotational systems", PGI-1, Forschungszentrum Jülich, Jülich, Germany, Dec. 4, 2024 (Seminar)
- "Unconventional torque in altermagnets induced by the magnetic octupole Hall effect", WE Heraeus-Seminar (Hybrid Angular Momentum Transport and Dynamics), Bad Honnef, Germany, Oct. 30, 2024 (Contributed talk)
- "Theoretical Calculations for Orbitronics: Orbital Hall Effect and Its Detection", Material Theory Division, Uppsala University, Uppsala, Sweden, Apr. 12, 2024 (Seminar)
- "First-principles calculation of the orbital current and orbital accumulation in metallic layers", DPG Spring Meeting 2024, Berlin, Germany, Mar. 21, 2024 (Contributed talk)
- "Theoretical Calculation of the Orbital Accumulation from the Orbital Hall Effect", Korean Magnetic Society 2023 Winter Conference, Busan, Korea, Nov. 23, 2023 (Contributed talk)
- "Theoretical calculations of the magneto-optical Kerr effect by the orbital Hall effect", PGI-1/IAS-1, Forschungszentrum Jülich, Jülich, Germany, May. 25, 2022 (Seminar)
- "Giant intrinsic orbital Hall effects in weakly spin-orbit coupled metals", American Physical Society March Meeting 2019, Boston, USA, Mar. 4–8, 2019 (Contributed talk)
- "Intrinsic Orbital Hall Effects in Elemental Solids", 11th BK21+ Young Physicists Workshop, Pohang, Korea, Feb. 14–15, 2019 (Contributed talk)

Poster

- "Unconventional orbital currents and torques via ferro-rotational order", The 14th International Conference on Advanced Materials and Devices (ICAMD 2025), Busan, Korea, Dec. 8–12, 2025
- "Unconventional orbital currents in ferro-rotational systems", SPEAR Conference on Spin-Orbitronics & 3rd Orbitronics Workshop, Donostia/San Sebastian, Spain, Mar. 31–Apr. 3, 2025
- "Current-Induced Orbital Current and Accumulation in Metallic Layers", 1st KAI-X Global Conference in Physics-International workshop on Orbitronics, Daejeon, Korea, Nov. 21, 2023
- "Current-Induced Orbital and Spin Accumulations in Metallic Layers", 11th International Symposium on Metallic Multilayers (MML 2023), Seoul, Korea, Jul. 24–28, 2023
- "Theoretical calculations of the magneto-optical Kerr effect by the orbital Hall effect", Orbitronics from topological matter to next level electronics (SPICE workshop), Ingelheim, Germany, Jul. 19–22, 2022
- "Orbital Hall effect and orbital torque in weakly spin-orbit coupled materials", Samsung Global Research Symposium, Seoul, Korea, Nov. 22–23, 2019
- "Giant Intrinsic Orbital Hall Effects in Light Metals", 15th SRC Winter Workshop on Topological Matter, Pohang, Korea, Jan. 30–31, 2019
- "Gigantic intrinsic orbital Hall effects in weakly spin-orbit coupled materials", International School on Spintronics and Korea-Japan Spintronics Workshop, Nagoya, Japan, Jan. 21–22, 2019

- "Giant intrinsic orbital Hall effects in weakly spin-orbit coupled systems", 2018 Korean Physical Society Fall Meeting, Changwon, Korea, Oct. 24–26, 2018